PVP 23

Code: 23CS3501

III B.Tech - I Semester - Regular Examinations - NOVEMBER 2025

DATA WAREHOUSING AND DATA MINING
(COMPUTER SCIENCE & ENGINEERING)

Duration: 3 hours Max. Marks: 70

Note: 1. This question paper contains two Parts A and B.
2. Part-A contains 10 short answer questions. Each Question carries 2 Marks.
3. Part-B contains 5 essay questions with an internal choice from each unit. Each Question carries 10 marks.
4. All parts of Question paper must be answered in one place.

BL – Blooms Level CO – Course Outcome

PART – A

		BL	CO
1.a)	Define Data Warehouse.	L1	CO1
b)	Explain with an example why data preprocessing is necessary.	L2	CO1
c)	List any two stages in the Machine Learning pipeline.	L1	CO1
d)	Calculate Euclidean distance to compute the distance between points (2, 3) and (5,7).	L2	CO2
e)	Define entropy in decision trees.	L1	CO2
f)	Draw a confusion matrix from the following results: TP=3, TN=4, FP=2, FN=1.	L2	CO2
g)	What is meant by linearly separable data?	L1	CO2
h)	Compare linear regression and logistic regression in two points.	L2	CO2
i)	List any two clustering algorithms.	L1	CO3
j)	Show one iteration of K-means clustering for the points (1,1), (2,1), (5,4) with K=2 and initial centroids C1= (1,1), C2= (5,4).	L2	CO3

PART – B

			BL	CO	Max. Marks
UNIT-I					
2	a)	Explain the architecture of a Data Warehouse with a neat diagram.	L2	CO1	5 M
	b)	Summarize the concept of Data Cube with examples of roll-up, slice and dice operations.	L2	CO1	5 M
OR					
3	a)	Discuss the importance of Data Cleaning, Integration and Transformation in preprocessing.	L2	CO1	5 M
	b)	Discuss the effectiveness of dimensionality reduction methods like PCA and attribute subset selection.	L2	CO1	5 M
UNIT-II					
4	a)	Distinguish between supervised learning and unsupervised learning with suitable examples.	L2	CO1	5 M
	b)	Describe the stages of a typical Machine Learning process with examples.	L2	CO1	5 M
OR					
5	a)	Point out the limitations of KNN compared to decision trees.	L4	CO4	5 M
	b)	Compare Machine learning with traditional programming with examples.	L2	CO1	5 M

UNIT-III					
6	a)	Explain the importance of pruning in decision trees and its effect on over fitting.	L2	CO2	5 M
	b)	Illustrate the role of Bagging and Boosting in improving classification accuracy.	L3	CO2	5 M
OR					
7		Construct a decision tree using ID3 algorithm for the dataset below to classify whether a loan will be approved: (L3, CO2)	L3	CO2	10 M

Income	Credit History	Collateral	Loan Approved
High	Good	Yes	Yes
High	Poor	No	No
Low	Good	Yes	Yes
Low	Poor	Yes	No
Medium	Good	No	Yes
Medium	Poor	No	No

UNIT-IV					
8	a)	Explain the limitations of perceptron and how SVM overcomes them.	L2	CO4	5 M
	b)	Apply Perceptron Learning Algorithm on the dataset below to separate the classes:	L3	CO2	5 M

X1	X2	Class
2	2	+1
4	4	+1
-1	-2	-1
-2	-3	-1

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<tr><td colspan="6">OR</td></tr>
<tr><td rowspan="2">9</td><td>a)</td><td>Compare linear regression and logistic regression with examples.</td><td>L4</td><td>CO4</td><td>5 M</td></tr>
<tr><td>b)</td><td>Using Logistic Regression, predict whether a student will pass the exam (Pass=1/Fail=0) for Hours_Study = 4:
<table>
<tr><th>Hours_Study</th><th>Pass</th></tr>
<tr><td>1</td><td>0</td></tr>
<tr><td>2</td><td>0</td></tr>
<tr><td>3</td><td>1</td></tr>
<tr><td>5</td><td>1</td></tr>
<tr><td>6</td><td>1</td></tr>
</table></td><td>L3</td><td>CO2</td><td>5 M</td></tr>
<tr><td colspan="6">UNIT-V</td></tr>
<tr><td rowspan="2">10</td><td>a)</td><td>Differentiate between Agglomerative and Divisive clustering.</td><td>L2</td><td>CO3</td><td>5 M</td></tr>
<tr><td>b)</td><td>Apply K-Means clustering with K=2 on the dataset below (initial centroids: C1= (1,1), C2= (5,5)). Show two iterations.
Points = {(1,1), (2,1), (1,4), (5,4), (6,5)}</td><td>L3</td><td>CO3</td><td>5 M</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td rowspan="2">11</td><td>a)</td><td>Illustrate Fuzzy C-Means clustering with a simple dataset.</td><td>L3</td><td>CO3</td><td>5 M</td></tr>
<tr><td>b)</td><td>Analyze the advantages and disadvantages of Expectation-Maximization (EM) clustering.</td><td>L4</td><td>CO4</td><td>5 M</td></tr>
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